Reaching Q3 in Formula 1 Across Regulation Eras: An RDD

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Abstract

Within Formula 1, for many mid/rear-field teams, reaching Q3 is a clear goal and objective. This is obvious, however, what is not is the points gain from reaching Q3 as opposed to marginally failing to reach Q3; this is the purpose of this project, to use the threshold around the cut-off time to qualify for Q3 and regression discontinuity to assess the causal effect of getting through to Q3, compared to not for the drivers/teams that are on the boarder of reaching Q3. Furthermore, I also attempt to assess how the qualifying tyre rule of having to start on the Q2 tire in the race if you reach Q3 impacted the points scored. This is done by differencing the discontinuities between the 2014-2021 era and the 2022-2025 era.
Using a RDD I found that there was a relatively large and statistically significant local effect of reaching Q3 for the 2022-2025 ground-effect era. However, for the 2014-2021 era, the result was near zero and statistically insignificant, suggesting that reaching Q3 had little causal effect on points, locally. Using this information, as well as some assumptions discussed later I show that the effect of the tire regulation for the 2014-2021 regulations negated the positive effect of getting into Q3 for a driver/team near the threshold for qualification.

1 Introduction

In 2025, the ground-effect era, beginning in 2022, came to an end, and with it a complete dataset for the era of the regulation was finished. These were also the years that the qualification regulations changed from the prior period where, had you surpassed the threshold for the 3rd round of qualifying (Q3), you would have to start the race on the tyres that you set your fastest lap in the 2nd round of qualifying (Q2) on. In the first round of qualifying all of the drivers take part (provided that they and the car are able to) and only the 15 fastest get through to the 2nd round of qualifying. Then there is Q2; this is where most of the empirical analysis for this project is focused. Q2 acts as the bridge between the first round and the final round, where the fastest 10 drivers will proceed to Q3, where their starting positions are decided (providing no penalties). When it comes to the race, only drivers that classify in the top 10 score points on a scoring system that is included in the appendix. The salient point is that for teams near the middle to back of the grid, even a single point can push them up the constructors championship, resulting in a large increase in prize money. This is due to the fastest teams filling out the top 10 nearly always. Therefore, it is valuable to assess the numerical impact of getting into Q3 for these teams that might regularly be near the threshold such that they can adapt their strategy in light of this causal effect. Moreover, assessing the impact that the tyre rules had on Q3's causal point change is useful for future FIA regulation considerations.

2 Theory and Setup

Towards the end of Q2 all teams within Q2, aside from the most confident, will go on a final push lap, this is to ensure that they are able to get through to Q3. Frequently the gap between the times set by the drivers in 10th and 11th are very small, so small that it is essentially random noise such as wind or slight variation in the track temperature between runs that decides their placement. Since those immediately either side of the threshold are separated by random noise, you can apply RDD to assess the causal effect of reaching Q3. This is valid due to there - on average - being no discernable features between the one directly north of the threshold and the one directly south, therefore, the point difference between the two must be down to essentially getting into Q3. However, if just using position in qualifying (i.e. p11 or p10), since it is discrete around the discontinuity, therefore, as supported by Zhuang (2026), to induce continuity, I defined the threshold for a given race r as

$$c_r = \frac{t_{(10)r} + t_{(11)r}}{2} \tag{1}$$

Since there are many Q2's where the gap between p11 and p10 in Q2 are near identical, an example being at Silverstone 2020 when Stroll and Gasly had identical Q2 lap times. The running variable is then stated to be the percentage gap between a given drivers lap time and the cut-off time following the formula below.

$$X_{ir} = 100 \cdot \frac{c_r - t_{ir}}{c_r} \tag{2}$$

This allows for a continuous running variable, as opposed to just having position which belongs to simply the natural numbers from 1 to 15.
Another potential issue is using points as the dependent variable, this is due to those below the threshold being unlikely to even score a single point, meaning that there is a potential lack of information. However, I believe that keeping points is still acceptable due to there being many cases where drivers do score after qualifying outside of the top 10, moreover, keeping points as the dependent variable gives an explicit value to getting through to Q3 for teams on the margin due to points being what decides the prize money, making the value of getting into Q3 more explicit.

The treatment effect (getting into Q3 for those near the threshold) for each era is defined as usually.

$$\beta_k = \lim_{x_{ik} \downarrow 0} \mathrm{E}[Y_{ik} | X_{ik} = x_{ik}] - \lim_{x_{ik} \uparrow 0} \mathrm{E}[Y_{ik} | X_{ik} = x_{ik}] \tag{3}$$

And since we assume continuity at $X_{ik} = 0$

$$\beta_k = \mathrm{E}[Y_{ik}(1) - Y_{ik}(0) | X_{ik} = 0] \tag{4}$$

Where $Y_{ik}(1)$ and $Y_{ik}(0)$ represents treated and non-treated drivers respectively. In reality, R calculated it using a triangular kernel.

In order to assess the impact of the 2014-2021 qualifying tire regulations, I assumed that the local causal effect of getting into Q3 for those near the threshold was the following.

$$\beta_{early} = \gamma_{early} + \alpha \tag{5}$$

$$\beta_{late} = \gamma_{late} \tag{6}$$

$$\implies \alpha = \beta_{early} - \beta_{late} + c \tag{7}$$

Such that α represents the effect of the tyre regulations. This means that the effect can be calculated by the difference of the two plus some constant that accounts for the difference between the pure Q3 benefit for each era γ_k. If if can be shown that c is approximately or at least close to 0, then it can be suggested that

$$\alpha \approx \beta_{early} - \beta_{late} \tag{8}$$

This is further addressed in the section 3, by showing that c is likely small by using average on-track overtakes by era and differencing the discontinuities in positions gained around the Q1 threshold for the two periods.

3 Empirical results

Figure 1 below shows the scatter plot of the race points achieved regressed against the running variable that is the percentage margin to Q2 cut-off, it can be seen that there is a clear jump at zero, suggesting that there is discontinuity, this is further supported by the RD effect having a p-value of 0.017, suggesting that there is sufficient evidence to reject H_0 of no RD effect at the 5% significance level. The estimated local effect is 1.57 points, this is relatively large, suggesting that the value of reaching Q3 for the marginal teams is - on average - an additional 1.57 points per race for a given driver.

There is more to discuss about this plot though, such as the collection of points below the regression line on the right-hand-side, suggesting that these are drivers that set a fast final Q2 time but scored lower points than drivers that set a final Q2 time slower than them. These are likely drivers belonging to top - but not the fastest - teams that needed to to a second run on a new set of tyres to make sure they made it through to Q3. This suggests that there is a second point where the regression breaks, approximately at the point past 1% positive of the cut-off. Despite this, it should not impact the analysis since it is sufficiently far away from the Q2 cut-off not to cause an issue as those just either side of the Q2 cut-off are likely similar, but those drivers mentioned before belong to top teams, not effecting the local estimate, as the teams on the margin of Q3 qualification will also be running again with a new set of tyres.

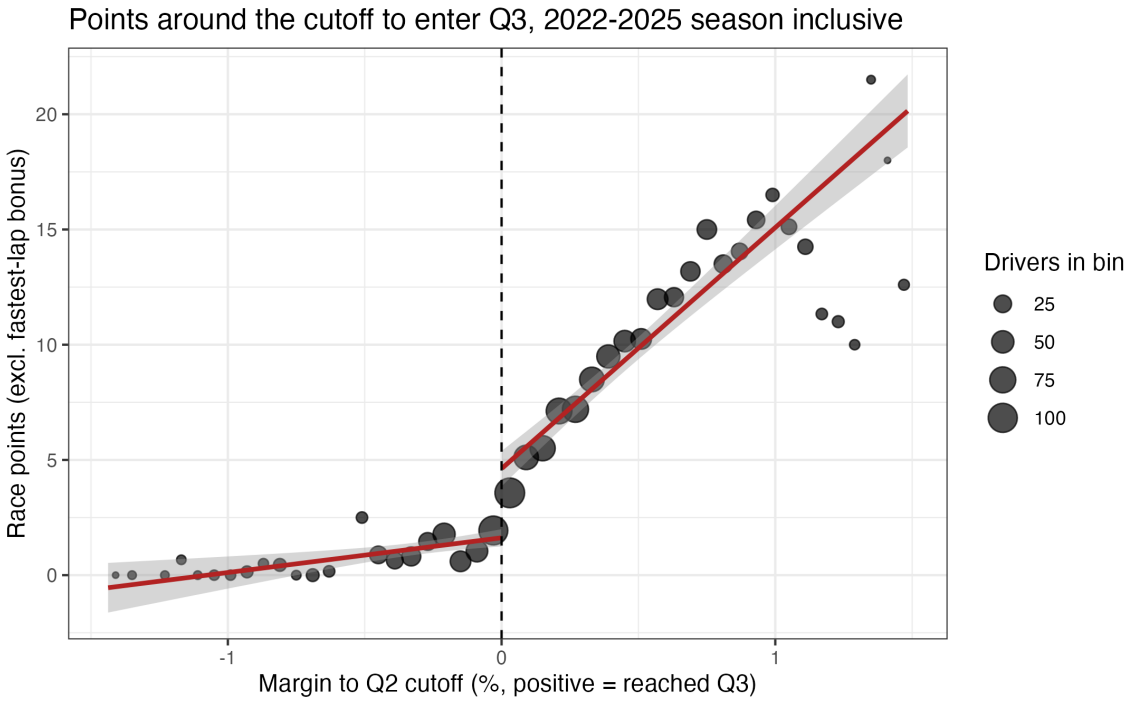


Figure 1: Scatter plot for 2022–2025 era. Data from Perry (2026).

Figure 3 is the scatter plot for the 2014–2021 era, where the jump appears to be non-existent despite there being a change in the gradient, stemming from the fact that a 1% change in pace below the cut-off will have a far smaller points effect than a 1% pace increase above the cut-off. This is supported by the estimated value being -0.139 points (small and negative) with a p-value of 0.542, thus, it is not statistically significantly different from zero. This is a key point, since for the 2022–2025 era, the estimated effect is strongly positive, however, reaching Q3 has no statistically significant local effect for the 2014–2021 era. As mentioned earlier, this could be down to the period's tyre qualifying regulations, where just reaching Q3 would force the driver to use their Q2 tyres (providing it isn't wet), which may not be strategically optimal, therefore, any single place advantage that that driver had over the driver that just missed Q3 may have been completely eliminated or made negative due to the suboptimal tyre being run.

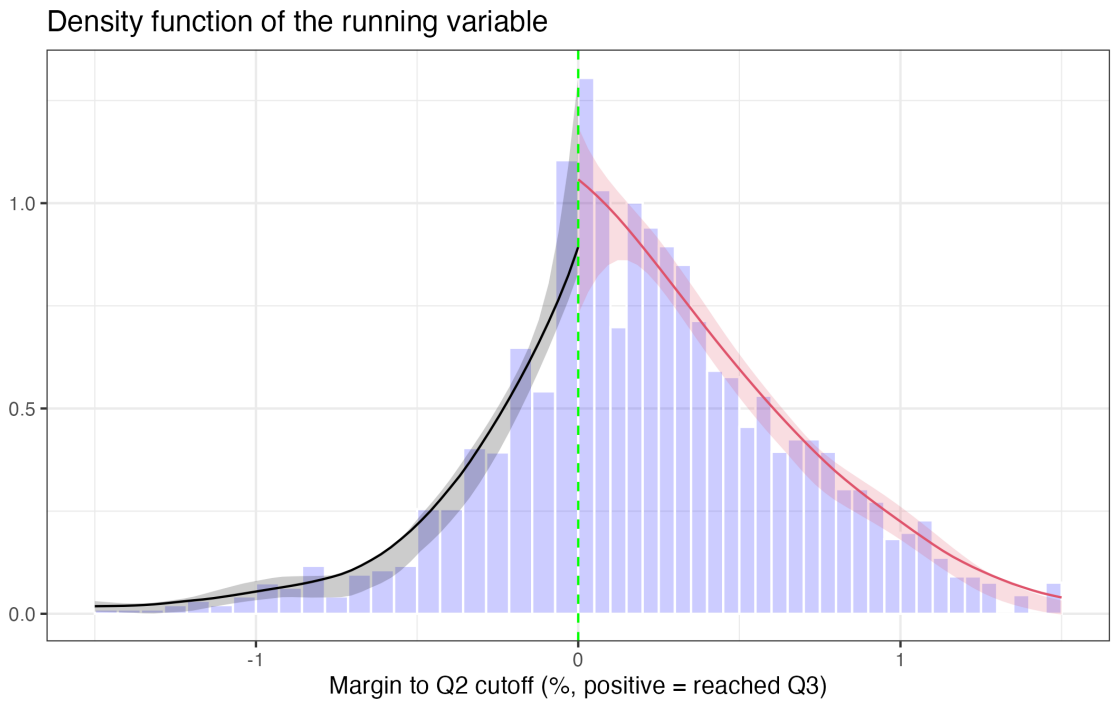


Figure 2: Density plot for 2022–2025 era. Data from Perry (2026)

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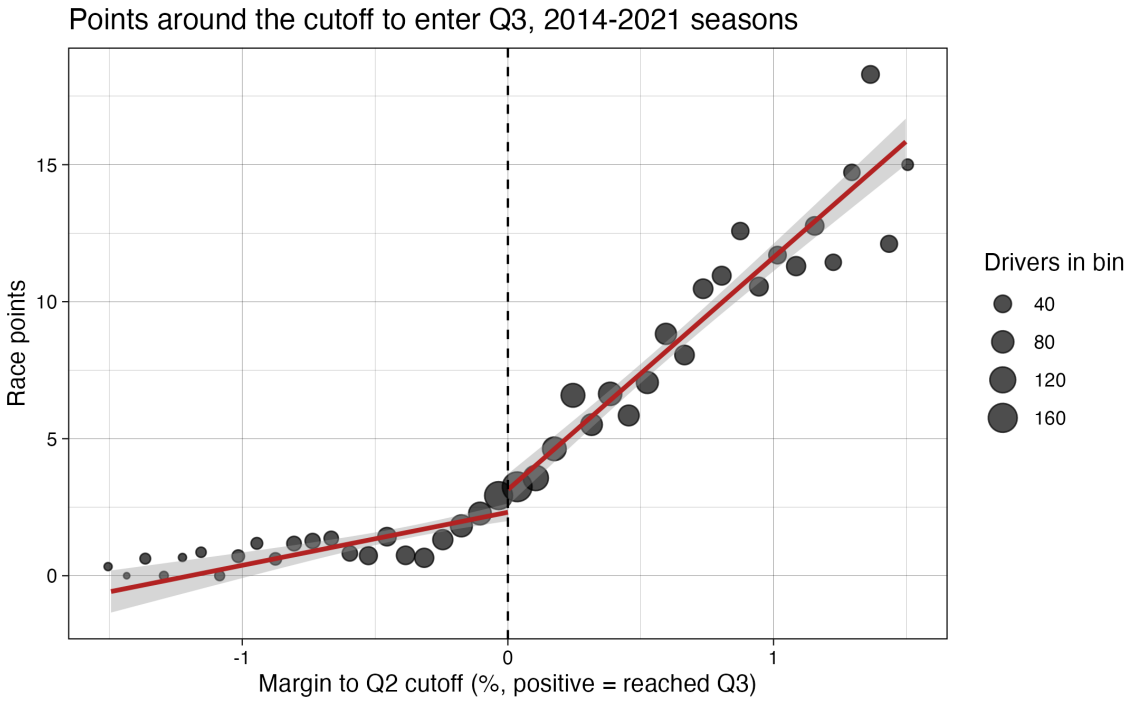


Figure 3: Scatter plot for 2014–2021 era. Data from Perry (2026)

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In order to make sure that it passes the falsification checks, I ran balance checks and assessed if sorting is present within the data. The balance check measures if there are any measured covariates that jump around the threshold (i.e. if a driver that just made it through to Q3 has any statistically significant Q1 differences between the driver that just missed out on Q3). The p-values for both eras' balancing checks in table 1 show that there is insufficient evidence to reject that there are no measured covariates that jump at the threshold, therefore, the treated drivers have no other treatment prior to reaching Q3.

Figure 2 and 4 show the plotted density of the running variable to see if sorting is a problem, which it could be given that some of the top teams may intentionally pace themselves to make sure that they just get through to Q3 without having to push the car and risk crashing or mechanical failure, moreover, in the 2014-2021 regulations, teams could intentionally position themselves to qualify p11, so that they don't have to run the Q2 tyre. Figures 2 and 4 in conjunction with the p-values in table 1 clearly show that there is no evidence of sorting being present just above the threshold, demonstrating that there is no large jump just to the right of the cut-off relative to just before the cut-off, instead, seeing a gradual rise to the model point which is around the cut-off.

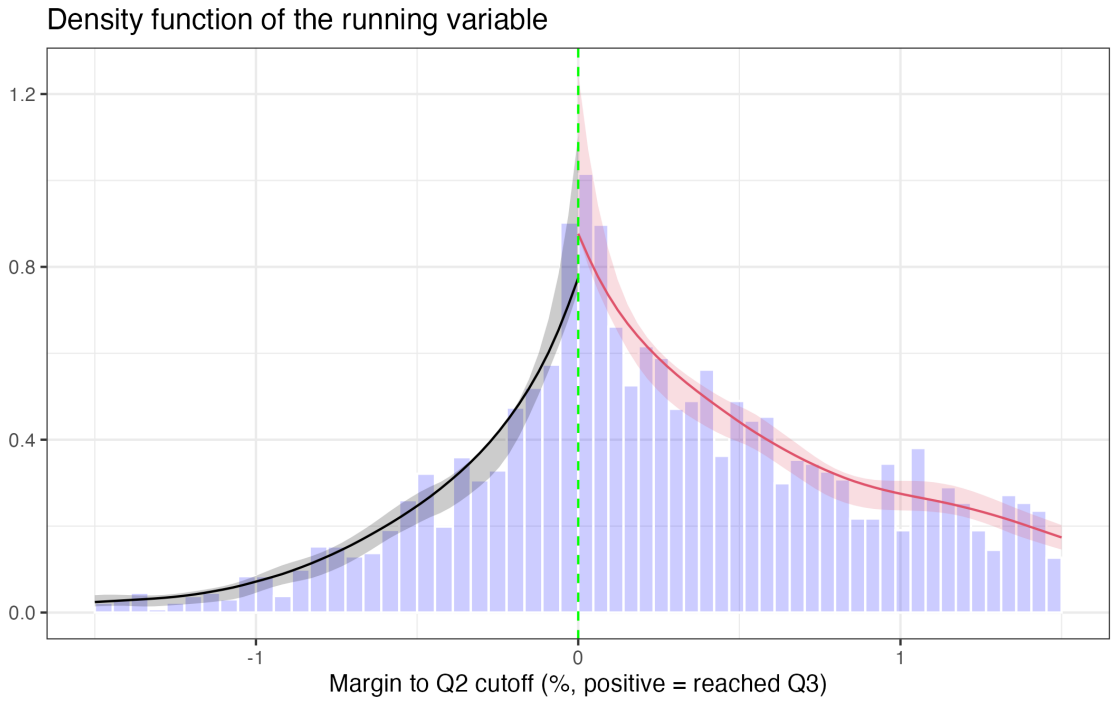


Figure 4: Density plot for 2014–2021 era. Data from Perry (2026)

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Era	RD Estimate	RD P-Value	Balancing Check P-Value	Sorting P-Value
2014-2021	-0.139	0.542	0.451	0.3315
2022-2025	1.570	0.017	0.986	0.533

Table 1: Estimates and P-values

In order to support that the benefit of track position hasn't changed, I used time series data to show that the number of overtakes by race is stationary, using a KPSS and ADF test for a unit root. Moreover, to make sure that there is no structural break around the beginning of the 2022 season, I also used a Zivot-Andrews Unit Root Test. Below is the plot of the time series showing the average number of overtakes per lap for each race between 2014-2024 (limited to 2024 due to the dataset ending there). Running ADF, KPSS, and a ZA demonstrate that the process is stationary and the most likely point of structural break if there is a structural break is not anywhere near the 2022 regulations, suggesting that overtaking is a stationary process moving around a constant mean for the whole time series. Given the large spike in 2016, I decided to rerun the ZA test excluding this outlier and the results held, as demonstrated in the table below.

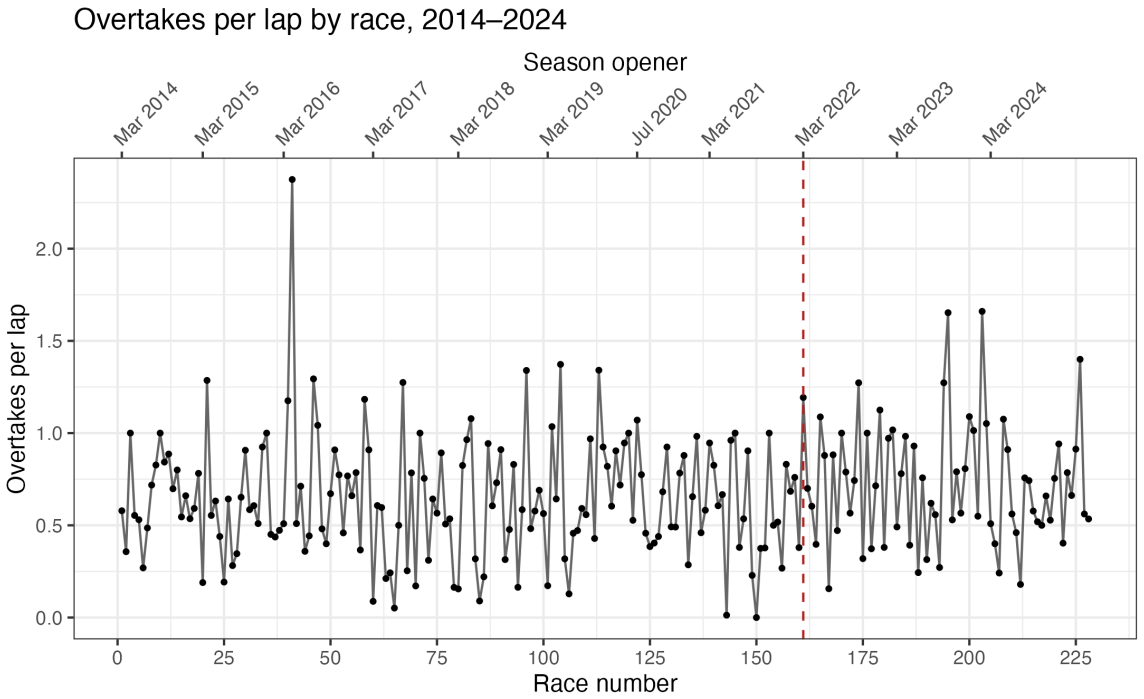


Figure 5: Time series for average overtakes per lap. Shows that it is stationary and no structural break. Red line illustrates beginning of 2022 regulations. Data from Rohanrao Vopani (2024).

Test	P-Value	Conclusion
ADF	0.01	Sufficient evidence to reject H_0 of unit root at 5% significance level
KPSS	0.1	Insufficient evidence to reject H_0 of stationarity at 5% significance level
ZA with outlier	0.01>	Sufficient evidence to reject H_0 of unit root with no structural break
ZA without outlier	0.01>	Sufficient evidence to reject H_0 of unit root with no structural break

Table 2: Table containing stationarity analysis for overtaking time series.

The data for figure 5 was generated from lap-by-lap data using an algorithm to calculate the number of on-track overtakes each lap, then averaged by dividing by the number of laps, this is done to prevent pit stops generating meaningless overtakes. From this, there is sufficient evidence to suggest that the process is stationary around a constant mean. Moreover, the break that the ZA does detect is the 58th race or the 21st round of the 2016 season, over five years prior to the regulation change, suggesting that there is no structural change in the number of overtakes per lap after the regulations came in compared to before.
Now, if you assume that the average number of overtakes per lap is a valid measure of the value of track position, i.e. the more overtakes there are the less track position matters due to positions being less secure, then you can assume that the value of track position in the 2014–2021 regulation is approximately the same in the 2022-2025 regulations. If this is the case then it implies that the value of making it through to Q3 locally (near the threshold) is approximately the same for both sets of qualifying regulations. This facilitates the use of difference in discontinuities to approximately

measure the effect of the qualifying tyre regulation on average points scored if you just pass the threshold for Q3. Given that the estimated local effect of reaching Q3 for the 2022–2025 is 1.570 and the estimated local effect of reaching Q3 for the 2014–2021 is -0.139, differencing the two leads to −1.709 being the approximate local points effect of the tire rule for the 2014-2021 era. It must be said, however, that using average overtakes per race is not a strong proxy and is only indicative as the mid-field (which we are concerned with) may suffer from a lack of overtaking whilst the front runners are perpetually swapping positions in one era and the reverse in the other; thus, there are potentially better measurements.

One such measure is using the Q1/Q2 cut-off regression discontinuity as a way to assess if there is a change in the value of grid positions between the two era's. This is done, not by using points - due to driver that qualify 16th/15th seldom receive points - but by subtracting their finishing position from their grid position, such that a positive number would indicate position(s) gained. This doesn't make the time series analysis redundant but works in conjunction with it.

Era	Estimate	P-Value
2014-2021	2.375	0.000
2022-2025	2.276	0.002

Table 3: Table containing estimates and p-values for the Q1/Q2 RDD.

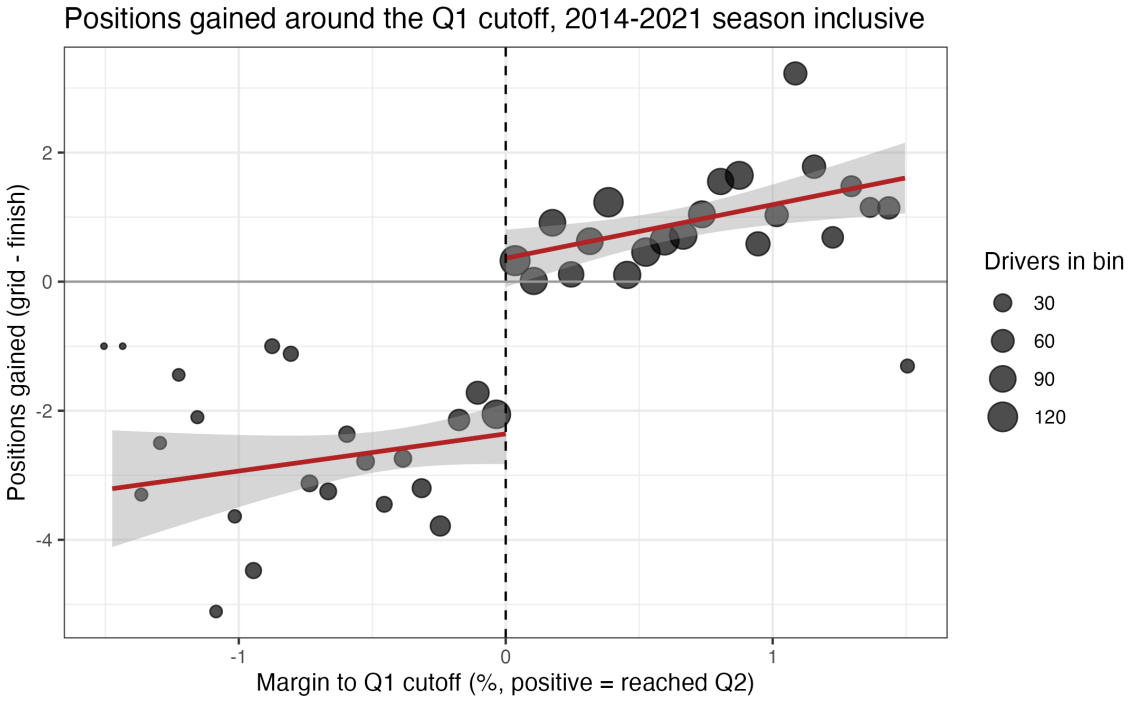


Figure 6: Scatter plot for 2014-2021 era positions gained regressed against margin to Q1 cut-off. Perry (2026).

Now, to make sure that the two are not statistically significantly different, I used a t-test, resulting

in a p-value of 0.9058, providing insufficient evidence to reject the null of them being equal to each other.

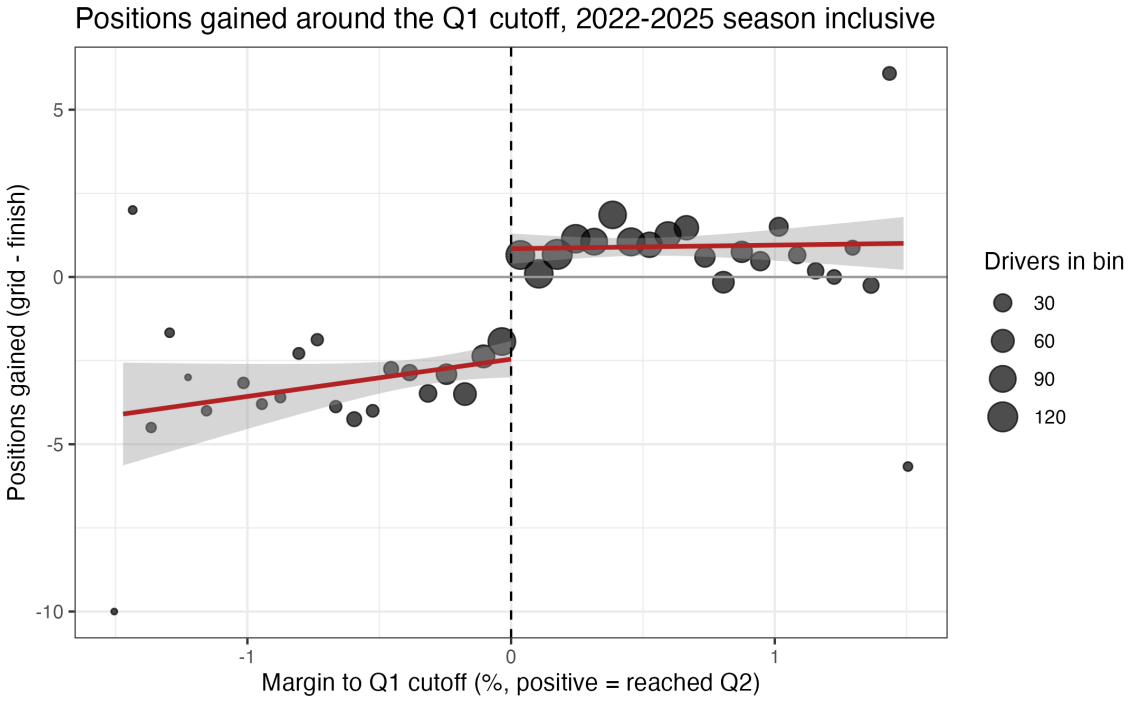


Figure 7: Scatter plot for 2022-2025 era positions gained regressed against margin to Q1 cut-off. Perry (2026).

Table 3 contains the estimates and the p-values for the RD estimates for the Q1/Q2 cut-off discontinuities, where it can be seen that they are both statistically significant and approximately the same, thus, supporting that the value of grid position is essentially the same across both eras. This in conjunction with the assessment of stationarity of over takes, supports that the effect of the tyre rule was approximately -1.709.

Estimate	P-Value	SE	95% CI
-1.709	0.030343	0.7892	[-3.256,-0.162]

Table 4: Estimate, p-value, standard errors, and confidence interval for the difference in discontinuities.

4 Conclusion

The main findings of this project is that, for the 2022–2025 era, the local effect of reaching Q3 was large and significant; 1.57 point - on average - represents a lot for teams in the mid to back of the field, such as Racing Bulls and Aston Martin being separated by only 3 points in the 2025 constructors championship, however, received $8 million USD difference (Phelan, 2026). This evidence may be interpreted as suggesting that teams should prioritise qualifying pace over race-pace, however, this

empirical study doesn't assess the cost of qualifying setup on race-pace. Moreover, the estimate is generated under prior team behaviour, if such behaviour were to change it is unlikely that the estimate would hold even controlling for regulations (hypothetically) especially since places in the top 10 are limited and this estimate only holds locally for drivers immediately either side of the Q2 cut-off. This acts as a potential example of Lucas' critique.

Additionally, this paper found that the likely local effect of the 2014-2021 tire regulations were largely negative, completely removing the benefit of reaching Q3 for mid to rear field teams. This finding could be beneficial in the future when motor sport governing bodies like the FIA attempt to change regulations, so that they can potentially understand the empirical impact of such regulations. Although this finding is key, it was built upon some strong assumptions. The first assumption was that you can treat the entire 2014-2021 period as a single era, where there were key regulation changes within. The second assumption was that the local casual effect of reaching Q3 was the same between each period. This is a very strong assumption, which is why it is stated that the two are only approximately the same, which is still somewhat strong but reasonable. I addressed this through differenced regression discontinuities for the Q1 cut-off between the two periods, where it was found that there is no statistically significant difference between the value of getting through to Q2 in the 2014–2021 era compared to the 2022–2025 era. This, paired with the 95% CI for the differenced discontinuities being all negative, suggests that at the very least - with a high degree of confidence - I can say that the Q1 tyre regulations did reduce the local causal benefit of surpassing the threashold to entre Q3.

References

Alex Kraieski, 2026. *Formula 1 data* [Online]. Available from: https://github.com/arkraieski/formula1data/blob/master/README.md [Accessed 1 October 2026].

Perry, J., 2026. *Jolpica/jolpica-f1-data* [Online]. original-date: 2025-06-15T08:35:23Z, 3 May. Available from: https://github.com/jolpica/jolpica-f1-data [Accessed 1 October 2026].

Phelan, M., 2026. *F1 prize money explained: how much do formula 1 teams and drivers really earn?* [F1 History] [Online], 19 June. Available from: https://www.formulaonehistory.com/f1-prize-money-explained-how-much-do-formula-1-teams-and-drivers-really-earn/ [Accessed 3 October 2026].

Rohanrao Vopani, 2024. *Formula 1 world championship (1950 - 2024)* [Online]. Available from: https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020 [Accessed 1 October 2026].

Zhuang, Z., 2026. *Regression discontinuity with cluster-level quotas: random score boundaries, finite rank margins, and an illustration from formula one qualifying* [Online]. 7006558. Rochester, NY: Social Science Research Network, 27 June. Available from: https://doi.org/10.2139/ssrn.7006558 [Accessed 3 October 2026].

A

Finishing Position	Points
1	25
2	18
3	15
4	12
5	10
6	8
7	6
8	4
9	2
10	1

Table 5: Points scored for given position excluding fastest lap which rewarded a single point if you were in the top 10 prior to the 2025 season.